

## HW 3 Solutions to non-graded problems

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a

Since  $\alpha, \beta, x_i$  are constants and the  $\epsilon_i$  are independent  $N(0, \sigma^2)$  variables, the  $Y_i$  are independent  $N(\alpha + \beta x_i, \sigma^2)$  variables. Therefore their joint likelihood function is

$$L_{x_1, \dots, x_n, \alpha, \beta, \sigma^2}(y_1, \dots, y_n) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2\right).$$

b

$$\partial_{\sigma^2} \log L = -n/(2\sigma^2) + (1/2\sigma^4) \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2 = 0$$

$$\implies \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta} x_i)^2,$$

$$\partial_{\alpha} \log L = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \alpha - \beta x_i) = 0$$

$$\implies n\alpha = \sum_{i=1}^n (y_i - \beta x_i) \implies \hat{\alpha} = \bar{Y} - \hat{\beta} \bar{x},$$

$$\partial_{\beta} \log L = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \alpha - \beta x_i) x_i = 0.$$

Substituting  $\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{x}$  gives

$$\sum_i (Y_i - \bar{Y} + \hat{\beta}(\bar{x} - x_i)) x_i = 0$$

$$\implies \hat{\beta} = \frac{\sum_{i=1}^n (\bar{Y} - Y_i) x_i}{\sum_{i=1}^n (\bar{x} - x_i) x_i}.$$

You can check that  $\sum_i Y_i (x_i - \bar{x}) = -\sum_i (\bar{Y} - Y_i) x_i$  and  $\sum (x_i - \bar{x})^2 = -\sum_i (\bar{x} - x_i) x_i$ , so

$$\hat{\beta} = \frac{\sum_{i=1}^n Y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

c

Since  $Y_i \sim N(\alpha + \beta x_i, \sigma^2)$ ,  $\mathbb{E}Y_i = \alpha + \beta x_i$ . So by linearity of expectation (remembering that  $x_i$  are constants)

$$\begin{aligned}
\mathbb{E}\hat{\beta} &= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) \mathbb{E}[Y_i] \\
&= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) (\alpha + \beta x_i) \\
&= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \left( \alpha \sum_{i=1}^n (x_i - \bar{x}) + \beta \sum_{i=1}^n (x_i - \bar{x}) x_i \right) \\
&= \beta \cdot \frac{\sum_{i=1}^n (x_i - \bar{x}) x_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \\
&= \beta.
\end{aligned} \tag{1}$$

Therefore

$$\mathbb{E}\hat{\alpha} = \mathbb{E}\bar{Y} - \bar{x}\mathbb{E}\hat{\beta} = \alpha + \beta\bar{x} - \bar{x}\beta = \alpha.$$

Showing that  $\hat{\sigma}$  is unbiased is more complicated....

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a

$$\hat{\alpha} = 0.06016, \hat{\beta} = 0.92063 \text{ (using R)}$$

c

$$0.1849254 \text{ (using R)}$$

4

The joint pmf for the sample is

$$f(x_1, \dots, x_n) = e^{-n\lambda} \frac{\lambda^{x_1 + \dots + x_n}}{x_1! \dots x_n!}.$$

The random variable  $Y$  is Poisson with parameter  $n\lambda$ , so  $P(Y = y) = e^{-n\lambda} (n\lambda)^y / y!$ . Therefore, since  $x_1 + \dots + x_n = y$ ,

$$\begin{aligned}
P(X_1 = x_1, \dots, X_n = x_n | Y = y) &= \frac{P(X_1 = x_1, \dots, X_n = x_n)}{P(Y)} \\
&= e^{-n\lambda} \frac{\lambda^{x_1 + \dots + x_n}}{x_1! \dots x_n!} \cdot e^{n\lambda} \frac{y!}{(n\lambda)^y} \\
&= \frac{y!}{n^y x_1! \dots x_n!},
\end{aligned} \tag{2}$$

which doesn't depend on  $\lambda$ .

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The joint pdf is

$$\begin{aligned}
f(x_1, \dots, x_n; \theta_1, \theta_2) &= (2\pi\theta_2)^{-n/2} \exp\left(-\frac{1}{2\theta_2} \sum_{i=1}^n (x_i - \theta_1)^2\right) \\
&= (2\pi\theta_2)^{-n/2} \exp\left(-\frac{1}{2\theta_2} \sum_{i=1}^n (x_i^2 - 2x_i\theta_1 + \theta_1^2)\right) \\
&= (2\pi\theta_2)^{-n/2} \exp\left(-\frac{1}{2\theta_2} \sum_{i=1}^n x_i^2 + \frac{\theta_1}{\theta_2} \sum_{i=1}^n x_i - \frac{n\theta_1^2}{2\theta_2}\right) \\
&= \phi(u_1(x_1, \dots, x_n), u_2(x_1, \dots, x_n), \theta_1, \theta_2) \cdot h(x_1, \dots, x_n),
\end{aligned} \tag{3}$$

where  $u_1(x_1, \dots, x_n) = \sum_{i=1}^n x_i$ ,  $u_2(x_1, \dots, x_n) = \sum_{i=1}^n x_i^2$ ,

$$\phi(u_1, u_2, \theta_1, \theta_2) = (2\pi\theta_2)^{-n/2} \exp\left(-\frac{1}{2\theta_2} u_2 + \frac{\theta_1}{\theta_2} u_1 - \frac{n\theta_1^2}{2\theta_2}\right),$$

and  $h(x_1, \dots, x_n) = 1$  (constant). Therefore, by definition,  $U_1 = \sum_{i=1}^n X_i$  and  $U_2 = \sum_{i=1}^n X_i^2$  are jointly sufficient statistics for  $\theta_1, \theta_2$ .

**b**

Recall  $(n-1)S^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$ . Therefore, given we know  $\bar{X}, S^2$ , we can find  $U_1, U_2$  by

$$U_1 = n\bar{X}, \quad U_2 = (n-1)S^2 + n\bar{X}^2.$$

Hence the transformation  $(U_1, U_2) \rightarrow (\bar{X}, S^2)$  is invertible, so  $(\bar{X}, S^2)$  are also jointly sufficient for  $\theta_1, \theta_2$ .

**c**

Any other pair that is an invertible function of  $U_1, U_2$  will work; e.g.  $Y_1 = U_1 + U_2$ ,  $Y_2 = U_1 - U_2$ .